

Cross-Modal Thermal Image Compression via RGB Side Information

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Abstract—High-resolution (HR) thermal imaging is essential in domains such as autonomous navigation and remote sensing, but its deployment is often limited by the cost of thermal sensors and the bandwidth required to transmit HR thermal data. This paper presents a cross-modal thermal image compression framework that leverages high-resolution RGB images as side information to enhance and efficiently encode low-resolution (LR) thermal inputs. Our method first applies a Guided Thermal Image Super-Resolution (GTISR) network to generate an initial HR thermal estimate from the LR thermal image and its aligned RGB counterpart. Rather than compressing the super-resolved output directly, we compute and encode only the residual using a learned image compression (LIC) model, while the LR thermal image is encoded separately using a standard codec such as Versatile Video Coding (VVC). Because the residual has substantially lower entropy, this two-stage approach significantly improves rate-distortion performance. Evaluated on the Perception Beyond the Visible Spectrum (PBVS) 2024 GTISR Challenge dataset, our framework achieves a 17.09% BD-rate reduction over a thermal-only baseline. These results demonstrate the benefit of integrating RGB side information into the compression pipeline and highlight the potential of cross-modal strategies for efficient thermal imaging under bandwidth constraints.

Index Terms—Thermal Image Compression, Super-Resolution, Residual Encoding, LIC, RGB Guidance, VVC, GTISR

I. INTRODUCTION

The operational capabilities of thermal imaging provide a robust alternative to conventional RGB cameras, particularly in scenarios where environmental conditions compromise the performance of visible-light sensors. By detecting emitted heat in the mid-to-longwave infrared spectrum, thermal sensors function independently of illumination, enabling effective operation in total darkness [1]. Their ability to penetrate visual obscurants, like smoke and fog, further extends their utility. Consequently, thermal imaging has become integral to a variety of applications where consistent performance is critical, including autonomous systems [2], [3], medical imaging [4], [5], and surveillance [6], as it provides reliable data when weather, shadows, or cluttered backgrounds compromise visual information.

A significant barrier to the broader implementation of thermal technology is the trade-off between sensor

cost and image resolution [7]. Affordable thermal sensors typically offer low spatial resolution, which can compromise the fine-grained detail necessary for precise analysis. While HR thermal images are available, their high cost often makes them impractical for widespread deployment. Computational methods, specifically super-resolution (SR) enhanced by deep learning, offer a viable path to overcome this hardware limitation. However, applying standard single-image SR algorithms directly to thermal data yields suboptimal results, as these images inherently lack the rich, high-frequency textural details needed for effective reconstruction.

To address the shortcomings of single-modality SR, GTISR utilizes a multi-modal data fusion strategy. This technique leverages an HR RGB image as a structural guide to enhance a corresponding LR thermal image. By transferring intricate details such as edges and textures from the visual guide to the thermal data, GTISR can reconstruct an HR thermal image that preserves both the original thermal information and fine spatial details. This approach effectively circumvents the need for expensive thermal hardware.

While GTISR successfully addresses the hardware limitations, its application creates a critical downstream challenge: managing the enormous resulting data volume. The $\times 16$ super-resolution at the core of our approach causes an explosive 256-fold increase in the data footprint of each image. For target applications such as unmanned aerial vehicles (UAVs) or edge-based monitoring systems, which operate under strict bandwidth and power constraints, transmitting or storing such a massive data load is entirely impractical. This necessitates a paradigm shift away from treating super-resolution and compression as separate, sequential tasks, requiring instead a co-designed framework where compression is integral to the image enhancement pipeline.

This work introduces a novel cross-modal compression framework that integrates guided super-resolution as a predictive coding step. We leverage a high-resolution RGB image, assumed to be available at both encoder and decoder, as correlated side information to create a high-quality thermal prediction via the MSFFCT network [8]. Our key insight is to treat this prediction as a

baseline and then apply a lossy compression algorithm exclusively to the residual error. This is highly efficient, as a successful prediction yields a sparse residual with significantly lower entropy than the image itself. Our compression backbone builds upon the Learned Image Compression (LIC) architecture. [9]. The final bitstream is composed of two parts: the original low-resolution thermal image compressed via a standard codec (VVC), and the compressed residual from our learned model. Our results demonstrate that this predictive, multi-modal compression strategy substantially outperforms traditional single-modality thermal compression across a range of bitrates.

II. RELATED WORK

Traditional image and video codecs such as JPEG2000, HEVC, and VVC have long served as the foundation for visual data compression. JPEG2000 uses wavelet-based coding with scalability and region-of-interest encoding [10]. HEVC and its successor VVC significantly improve compression efficiency using advanced motion prediction, transform coding, and in-loop filtering [11], [12]. In our work, we adopt VVC to compress LR thermal inputs, benefiting from its efficient intra-frame coding. Unlike conventional methods that compress full-resolution images, we encode the LR thermal with VVC and apply learned residual compression to the enhanced SR content.

In the context of image compression, Ballé et al. introduced an end-to-end trainable autoencoder for LIC, which jointly optimized rate and distortion metrics [13]. This was extended with a scale hyperprior model to better capture spatial correlations in the latent representation, enabling improved entropy modeling [14]. Minnen et al. further enhanced LIC performance by combining autoregressive and hierarchical priors [15]. Although highly accurate, these methods often suffer from serial decoding inefficiencies. To mitigate this, Minnen and Singh proposed channel-wise autoregression [16], and He et al. introduced a checkerboard context model to parallelize decoding with minimal performance loss [17].

He et al. later proposed ELIC (Efficient Learned Image Compression), which balances speed and rate-distortion performance using channel grouping and lightweight network modules [9]. ELIC forms the basis for our residual compression, where computational efficiency and scalability are key.

Beyond RGB imaging, learned compression has been explored for other sensing modalities. Maharjan et al. introduced a neural codec for complex-valued SAR images, jointly encoding amplitude and phase for end-to-end optimization [18]. While SAR and thermal imaging differ, this work demonstrates the adaptability of LIC across non-RGB domains.

Ayzik and Avidan proposed a decoder-guided LIC scheme inspired by distributed source coding, which leverages a correlated reference image only available at the decoder [19]. Their system achieved substantial bandwidth savings, validating the benefit of utilizing cross-image redundancy—similar in spirit to our RGB-guided residual compression.

Finally, Bhowmik et al. analyzed lossy compression for thermal imagery, showing that even mild JPEG compression degrades the performance of the CNN detector, especially for small objects [20]. They also found that incorporating RGB guidance improves performance. These findings support our design choice of using RGB as high-resolution side information in thermal image reconstruction.

III. METHODOLOGY

To reconstruct a high-resolution (HR) thermal image from a low-resolution (LR) thermal input, we adopt the GTISR model based on the MSFFCT architecture [8]. Designed for large upscaling factors (e.g., $\times 16$), this model leverages HR RGB guidance to recover fine thermal details.

An overview of the proposed super-resolution and compression pipeline is shown in Fig. 1.

Let $\hat{I}_{\text{th}}^{\text{LR}} \in \mathbb{R}^{H \times W}$ denote the decoded low-resolution (LR) thermal image obtained from the VVC bitstream, and let $I_{\text{RGB}}^{\text{HR}} \in \mathbb{R}^{sH \times sW \times 3}$ be the corresponding high-resolution (HR) RGB image, where s is the upscaling factor. The decoded LR thermal image is first upsampled using bicubic interpolation:

$$\tilde{I}_{\text{th}}^{\text{HR}} = \text{Bicubic}(\hat{I}_{\text{th}}^{\text{LR}}). \quad (1)$$

The upsampled thermal image is concatenated with the HR RGB image:

$$I_{\text{cat}} = \text{Concat}(\tilde{I}_{\text{th}}^{\text{HR}}, I_{\text{RGB}}^{\text{HR}}). \quad (2)$$

The combined input is passed through a preprocessing block that applies pixel unshuffle operations to generate two reformatted representations:

$$F_{\times 2} = \text{Unshuffle}_{\times 2}(I_{\text{cat}}), \quad F_{\times 4} = \text{Unshuffle}_{\times 4}(I_{\text{cat}}). \quad (3)$$

These are processed by a shallow feature extractor composed of parallel convolutional layers:

$$F_{\times 2}^{(0)}, F_{\times 4}^{(0)} = f_{\text{shallow}}(F_{\times 2}, F_{\times 4}). \quad (4)$$

The reformatted features are then fed into a stack of Fusion Blocks. Each Fusion Block processes both $F_{\times 2}^{(0)}$ and $F_{\times 4}^{(0)}$ through parallel residual branches. The $\times 4$ path is upsampled using a transposed convolution (TConv) operation to match the spatial resolution of the

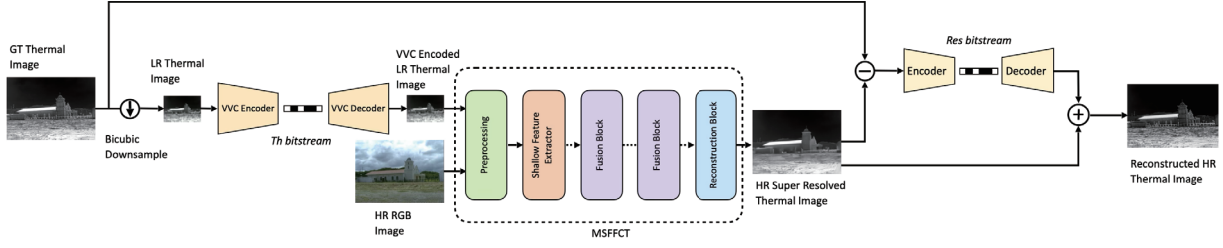


Fig. 1: Overview of the proposed super-resolution thermal image compression pipeline. The LR thermal image and HR RGB image are first processed by the MSFFCT block, in which the shared feature extractor and fused using a transformer-based fusion block to generate a super-resolved thermal image. The residual between the SR output and the ground truth HR thermal image is computed and compressed using a learned image codec. At the decoder side, the residual is decoded and added back to the SR thermal image to reconstruct the final high-quality thermal output.

$\times 2$ stream. The two are concatenated and passed through a channel-wise self-attention mechanism:

$$F_{\text{fus},1} = f_{\text{fusion}}^{(1)}(\text{Concat}(\text{Res}_{\times 2}, \text{TConv}(\text{Res}_{\times 4}))), \quad (5)$$

$$F_{\text{fus},i} = f_{\text{fusion}}^{(i)}(F_{\text{fus},i-1}), \quad \text{for } i = 2, \dots, 48. \quad (6)$$

Here, $f_{\text{fusion}}^{(i)}$ denotes the i -th Fusion Block, which integrates three core operations: residual refinement applied in parallel to the $\times 2$ and $\times 4$ feature streams, transposed convolution to spatially align the $\times 4$ output with the $\times 2$ stream, and channel-wise self-attention over the concatenated representation to model long-range and cross-modal dependencies.

The output of the final Fusion Block is passed through a reconstruction block, which upsamples the features and adds back the bicubic estimate:

$$\hat{I}_{\text{SR}} = f_{\text{recon}}(F_{\text{fus},48}) + \hat{I}_{\text{Th}}^{\text{HR}}. \quad (7)$$

The result $\hat{I}_{\text{SR}} \in \mathbb{R}^{sH \times sW}$ is the super-resolved thermal image. While this output restores most structural details, some high-frequency components may still be missing. To address this, we introduce a residual compression stage.

We compute the residual between the ground-truth HR thermal image and the super-resolved output:

$$r = I_{\text{GT}} - \hat{I}_{\text{SR}}. \quad (8)$$

Since r contains only the missing high-frequency information, it has lower entropy and is more suitable for compression.

We employ the Learned Image Compression (LIC) model [16] to encode the residual. LIC is a deep autoencoder with a factorized entropy model. It simulates quantization using additive uniform noise and compresses the latent representation via entropy coding. The model is trained with a rate-distortion loss:

$$\mathcal{L} = R + \lambda D, \quad (9)$$

where R is the estimated bitrate, D is the distortion (e.g., MSE), and λ governs the trade-off.

At the decoder, the LR thermal image is passed through the MSFFCT-based GTISR model using the HR RGB image to produce $\hat{I}_{\text{SR}}$. Simultaneously, the compressed residual is decoded using LIC to yield $\hat{r}$. The final thermal output is obtained by:

$$\hat{I}_{\text{final}} = \hat{I}_{\text{SR}} + \hat{r}. \quad (10)$$

This two-stage framework combines structure-preserving guided super-resolution with precise residual correction. By transmitting only the LR thermal image and a compressed residual, assuming HR RGB availability at the decoder, we achieve HR thermal reconstruction at significantly reduced bitrates.

IV. EXPERIMENTS AND RESULTS

A. Dataset and Setup

We evaluate our approach on the PBVS 2024 GTISR Challenge dataset [21]. This dataset, referred to as CIDIS, consists of 1000 pairs of co-registered visible and thermal images captured using a Basler visible camera and a FLIR Tau2 thermal camera. All images are spatially aligned with a resolution of 640×480 , making this dataset well-suited for cross-modal super-resolution tasks.

For model training, we use 700 image pairs for training and 100 for validation. The best-performing model, selected based on PSNR on the validation set, is used to generate super-resolved outputs for the compression stage. We evaluate our pipeline on 5 representative test images from the challenge set.

LR thermal inputs are generated by downsampling the ground-truth HR thermal images by a factor of $\times 16$, following the PBVS 24 dataset. These downsampled thermal images are compressed using the VVC reference software, employing intra-frame coding at a high compression ratio. The encoder is configured to

TABLE I: Rate-Distortion Comparison (Test Set)

Method	λ	BPP	PSNR (dB)
Proposed Method	0.15	0.0495	34.90
	0.5	0.0872	37.92
	1.5	0.1476	40.52
	3.0	0.2028	41.99
Thermal-Only Baseline	0.15	0.0690	35.62
	0.25	0.0850	36.82
	1.0	0.1533	39.64
	2.0	0.2069	40.74

produce a compressed thermal base that retains global structure, while finer details are recovered from the learned residual stream.

B. Implementation Details

The GTISR model is trained from scratch using paired RGB images and their corresponding LR thermal images obtained from VVC decoding. The training strategy follows that of the original MSFFCT model [8], where random 16×16 crops are extracted from the LR thermal images and their aligned HR RGB counterparts. The model is optimized using a composite loss function comprising L1 loss, structural similarity (SSIM) loss, and perceptual loss computed from VGG features. Data augmentation, including horizontal and vertical flips and 90° rotations, is applied during training.

For compression, we adopt the LIC framework, which consists of an analysis transform (encoder CNN), a quantizer, and a synthesis transform (decoder CNN), along with a hyperprior that models entropy parameters [14], [16]. The LIC network operates on residual images computed as the difference between the super-resolved prediction and the ground-truth HR thermal image. Residuals are randomly cropped into 256×256 patches to enable memory-efficient training.

We train four separate LIC models using different values of the rate-distortion trade-off parameter, with $\lambda \in \{0.15, 0.5, 1.5, 3\}$, covering a spectrum of operating points from low bitrate to high reconstruction accuracy. During inference, the model corresponding to the desired bitrate or quality level is selected. LIC models are trained until the BPP-PSNR performance plateaus in the validation set.

C. Quantitative Results

We evaluate rate-distortion performance using two standard metrics: bits per pixel (BPP) and peak signal-to-noise ratio (PSNR). BPP is computed as the total number of transmitted bits divided by the number of pixels in the reconstructed HR thermal image. PSNR (in dB) is measured between the reconstructed and ground-truth HR thermal images.

We compare our method against two baselines. The first, Thermal-Only, directly compresses the ground-truth HR thermal image using the same LIC codec

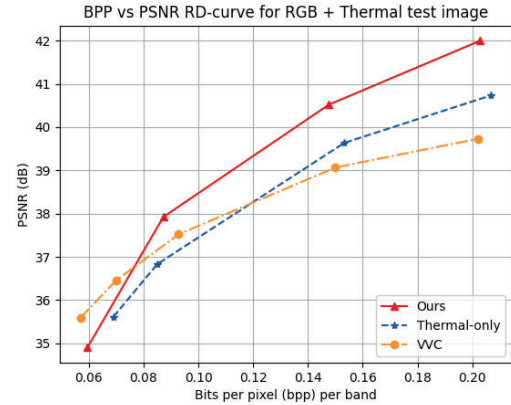


Fig. 2: Rate-Distortion (RD) performance comparison with the proposed method (Ours), the thermal-only baseline (Thermal-only) and VVC compression. Results are shown on the PBVS 2024 GTISR test image, demonstrating that our approach consistently achieves higher PSNR at lower bits-per-pixel (bpp), highlighting the advantage of leveraging RGB guidance and residual encoding.

without RGB guidance or residual learning. The second baseline uses the VVC reference software to intra-code the HR thermal image, representing a traditional video codec. Table I reports results across four λ values, each corresponding to a separately trained LIC model. Since both methods involve independently trained networks at different operating points, we match each proposed result with the closest baseline entry in terms of BPP for a fair comparison.

Our method consistently outperforms the thermal-only baseline across all bitrate settings. For example, at $\lambda = 3.0$, the proposed model achieves 41.99 dB at 0.2028 bpp, compared to 40.74 dB at 0.2069 bpp for the baseline. At $\lambda = 1.5$, the gain is nearly 0.9 dB at a lower bitrate. In the medium bitrate regime ($\lambda = 0.5$), our method reaches 37.92 dB versus 36.82 dB from the baseline. Even in the low-bitrate regime ($\lambda = 0.15$), although the baseline achieves a marginally higher PSNR (35.62 dB), it does so with 39% more bits than our model (0.0690 bpp vs. 0.0495 bpp).

These trends are illustrated in Fig. 2, which plots the rate-distortion curves for the proposed method and both baselines. The proposed framework consistently yields higher reconstruction quality at lower bitrates. This improvement is attributed to the RGB-guided prediction, which captures structural content effectively, leaving a lower-energy residual that is easier to compress.

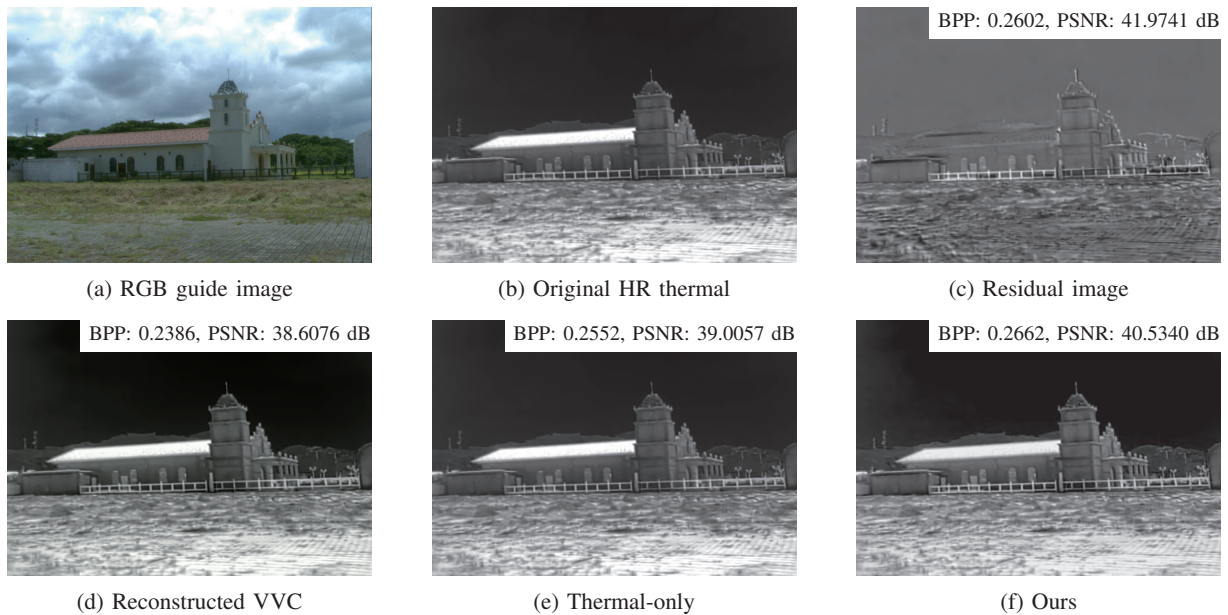


Fig. 3: Visualization of the thermal image pipeline: (a) HR RGB guide image, (b) ground-truth HR thermal image, (c) residual between ground truth and super-resolved prediction, (d) reconstructed VVC image, (e) reconstructed thermal-only image, and (f) our reconstructed image.

D. Qualitative Results

We further evaluate visual quality through qualitative inspection, as shown in Fig. 3. The proposed RGB-guided compression framework consistently produces images with sharper edges and more faithful texture preservation compared to traditional compression baselines. In particular, our method retains structural detail around fine contours, such as building edges and rooftop textures, which appear noticeably smoother or blurred in both the VVC and Thermal-Only reconstructions.

For instance, the reconstructed image in Fig. 3(f) demonstrates a higher visual quality than the ground-truth HR thermal image in Fig. 3(b), while operating at a comparable bitrate (0.2662 bpp vs. 0.2552 bpp in (e)). Moreover, the residual-based approach mitigates intensity mismatches and recovers finer thermal gradients, correcting for subtle errors introduced during the super-resolution stage. This leads to improved consistency in both low-frequency and high-frequency regions.

Additionally, our method outperforms the VVC-based baseline (Fig. 3(d)), which exhibits visible compression artifacts and edge degradation, especially in high-contrast areas. The residual image in Fig. 3(c) confirms that the RGB-guided prediction absorbs much of the structural content, leaving only minimal high-frequency correction, which is efficiently captured by the learned residual encoder.

These results support our key hypothesis: RGB guid-

ance combined with residual learning significantly enhances thermal image reconstruction quality under tight bitrate constraints.

E. Discussion

The results demonstrate the effectiveness of combining RGB-guided super-resolution with residual compression. Across medium and high bitrates, our method consistently outperforms direct thermal compression in terms of both PSNR and bitrate efficiency. This advantage stems from the RGB image providing a strong prior, enabling the SR network to reconstruct much of the thermal structure before compression. As a result, the entropy of the residual is significantly reduced, leading to more efficient coding. The network's contribution does not incur bitrate, allowing the codec to focus only on fine-grained residual correction.

At the lowest tested bitrate ($\lambda = 0.15$), the thermal-only baseline achieves slightly higher PSNR, likely because the limited bit budget prioritizes essential structures in a single-stage codec. In contrast, our two-stage pipeline may produce fine details in the SR image that cannot be fully corrected when the residual is heavily quantized. This suggests a bitrate threshold below which the residual correction is less effective, and a simpler interpolation strategy may suffice. However, at all higher bitrates, our approach consistently yields better rate-distortion trade-offs, showing clear gains once residual information can be adequately encoded.

From a deployment standpoint, the added computational cost of the SR and compression networks can be offset by modern GPU-accelerated inference. In scenarios like UAV thermal video or satellite imaging—where bandwidth is highly constrained—the proposed framework provides significant bitrate savings with minimal loss in fidelity. While our dataset assumes perfect RGB-thermal alignment, the method is extensible to less controlled settings by incorporating multi-modal registration or using alignment-aware architectures. These findings underscore the practical viability of RGB-guided residual compression for thermal imaging systems.

V. CONCLUSION

In this paper, we propose a cross-modal thermal image compression framework that exploits RGB side information to improve both reconstruction quality and bitrate efficiency. Our method combines RGB-guided super-resolution with learned residual compression, enabling the LR thermal image to be enhanced through the HR RGB counterpart, followed by compact encoding of only the residual content. Evaluated on the PBVS 2024 GTISR dataset [21], our approach consistently outperformed thermal-only compression baselines across a range of bitrates. Notably, it achieved up to 17.09% BD-rate reduction while preserving or improving visual quality. These results validate the effectiveness of integrating side information into the compression pipeline and highlight the potential of cross-modal coding strategies in bandwidth-constrained thermal imaging applications.

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